

॥ भगवान् श्री शमकुण्डा शरणम् ॥

Notes written by

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Generative Models:

examples: Chat GPT
Gemeni
Claude } CondⁿL text generators.

Generate op based on prompt

i/p: NL
Code

Image generators: DALLe
Stable diffusion } CondⁿL Image gen

Speech generators: Text $\rightarrow$ wav file / speech

Starting point

$D = \{x_1, x_2, \dots, x_n\} \sim \text{iid } P_X \text{ (unknown)}$

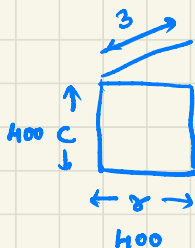
$x_i \in \mathbb{R}^d$, d -dimensionality of the data.

eg: x_i Images

$x_i \in \mathbb{R}^{r \times c \times 3}$

480000

$x_i \in \mathbb{R}$



X : Random Variable with a distribution P_X

→ X_i & X_j are statistically independent

→ we are not saying k^{th} dim & l^{th} dim of X_i are independent
(pixels are not independent)

→ X_i : instances of a vector valued RV of size d .

Generative Modeling:

Given $D = \{X_1, \dots, X_n\} \sim \text{iid } P_X$

Goal : Estimate P_X & learn to sample from it

* in Discriminative take $P_{Y|X}$ we learn the dist
& not sample from it.

General Principle of Gen Models:

→ Assume a parametric family on p_X (density funⁿ is well defined)
denoted by p_θ

p_θ : Represented using DNN
(Model)

2) Define and estimate a divergence metric

b/w p_0 & p_x

3) Solve an optimization problem over the parameters of p_0 to minimize the above divergence metric.

There is a RV $Z \in \mathbb{R}^k$ & has some arbitrary but known distribution

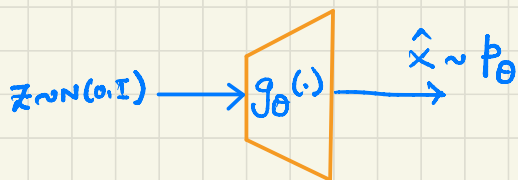
$$Z \sim N(0, I)$$

zero mean
Unit Variance.

Suppose $g_0(Z) : Z \rightarrow X$ then

$\hat{X} = g_0(Z)$ has a diff dist than
that of Z & dist of $\hat{X}$
depends on the func $g_0(\cdot)$

→ Suppose $g_0(Z)$ is a NN.



$$\hat{X} = g_0(Z)$$

Denote the density of $\hat{X}$
using $p_0(\cdot)$

* Suppose $D(p_x \parallel p_\theta)$ denote the divergence b/w
 p_x & p_θ , $D \geq 0$

* Solve the following optimization problem

$$\theta^* = \underset{\theta}{\operatorname{argmin}} D(p_x \parallel p_\theta)$$

* $D = 0$ iff $p_x = p_\theta$

* Upon solving the above optimization problem,

the distribution is implicitly estimated by $g_\theta(z)$

& one can sample from p_x using $g_\theta(z)$

* Sample from $z \sim N(0, I)$

pass through $g_\theta(z)$ would produce a sample
from p_{θ^*} which is close to p_x because of
solving optimization problem

The questions:

- 1) how to compute the divergence metrics without knowing p_θ & p_x , all we have are samples from both
 - 2) what should be the choice of the divergence metric?
 - 3) how to choose the $g_\theta(z)$, in turn p_θ ?
 - 4) how to solve the optimization problem of minimizing the divergence metrics?
-

Variational Divergence minimization:

Define divergence metrics b/w distributions.

f - Divergence:

given 2 prob dist func with the corresponding density functions denoted by p_x & p_0 , the f-divergence b/w them is defined as:

$$D_f(p_x \parallel p_0) = \int_X p_0(x) f\left(\frac{p_x(x)}{p_0(x)}\right) dx$$

Cond's: RV are continuous & density is defined

$$f(u) : \mathbb{R}^+ \rightarrow \mathbb{R}$$

Convex

Left Semi Continuous

$$f(1) = 0$$

X: Space on which the p_x & p_0 are supported

* Properties of f-divergence

1) $D_f(\cdot) \geq 0$ for any choice of $f(\cdot)$

2) $D_f(\cdot \parallel \cdot) = 0$ only when both the dist are same

Examples of f -Divergence:

1) $f(u) = u \log u$: KL-Divergence

$$\begin{aligned} D_{KL}(p_x \parallel p_\theta) &= \int_X p_\theta(x) \left[\frac{p_x(x)}{p_\theta(x)} \log \frac{p_x(x)}{p_\theta(x)} \right] dx \\ &= \int_X p_x(x) \log \left(\frac{p_x(x)}{p_\theta(x)} \right) dx. \end{aligned}$$

$$\underbrace{D_{KL}(p_x \parallel p_\theta)}_{\text{Forward KL}} \neq \underbrace{D_{KL}(p_\theta \parallel p_x)}_{\text{Reverse KL}}$$

2) $f(u) = \frac{1}{2} \left(u \log u - (u+1) \log \left(\frac{u+1}{2} \right) \right)$

JS divergence

3) $f(u) = \frac{1}{2} |u-1|$: TV distance

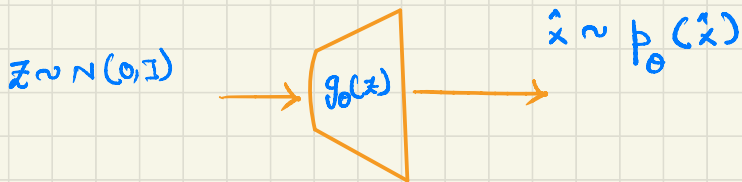
→ diff choice of f -divergence, Diff generative model

Algorithm for D_f Minimization:

Recall

$$D \sim \text{iid } p_x$$

Goal: Sample from p_x



The above setup becomes a sampler for p_x
if $p_\theta = p_x$

$$\theta^* = \underset{\theta}{\operatorname{argmin}} D_f(p_x \parallel p_\theta)$$

Objective: Algorithm to minimize D_f b/w p_x & p_θ
without knowing both of them

But having samples from both

Samples from p_x : D

Samples from p_θ : outputs of $g_0(z)$ for diff z

problem: how to integrate over a d -dim space
with samples

Key Idea:

Integrals involving density func can be approx using Samples drawn from the dist.

Suppose we want to compute

$$I = \int_{\mathcal{X}} h(x) p_{\mathcal{X}}(x) dx \quad h(\cdot) \text{ is a func on RV}$$

we have samples drawn iid from $p_{\mathcal{X}}$

$$\{x_1, \dots, x_n\} \sim \text{iid } p_{\mathcal{X}}$$

$$I = \mathbb{E}_{p_{\mathcal{X}}} [h(x)] \quad ; \text{ LOTUS}$$

Now by Law of Large Numbers,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n h(x_i) \approx \mathbb{E}_{p_{\mathcal{X}}} [h(x)]$$

$\therefore$ If the f-div can be expressed in terms of expectations of func wrt $p_{\mathcal{X}}$ & $p_{\mathcal{Y}}$, then One can compute & optimize them

Expressing D_f in terms of Expectations over p_x & p_0

$$D_f(p_x \| p_0) = \int_x p_0(x) f\left(\frac{p_x(x)}{p_0(x)}\right) dx$$

$$\int_x p_0(x) h(x) dx \neq E[.] \quad \text{since this is not instance of RV}$$

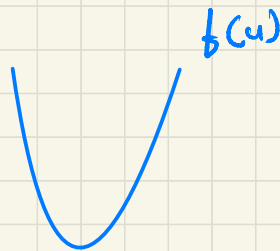
Conjugate func for a Convex func

If $f(u)$ is a convex func, then there exists a conjugate func $f^*(t)$ defined as follows:

$$f^*(t) = \sup_{u \in \text{dom}_f} \{ ut - f(u) \} \quad (\text{max})$$

This is a point wise func:

need to solve this @ every point



for any given point, get a lot of lower bounds and choose the tightest of them.

Properties of the Conjugate $f^*(t)$

1) $f^*(t)$ is also Convex

$$2) [f^*(t)]^* = f(u)$$

$$f(u) = \sup_{t \in \text{dom } f^*} \{ tu - f^*(t) \}$$

How is this related to f -divergence.

$$D_f(p_x \| p_\theta) = \int p_\theta(x) f\left(\frac{p_x(x)}{p_\theta(x)}\right) dx.$$

$$= \int p_\theta(x) f(u) dx \quad u = \frac{p_x(x)}{p_\theta(x)}.$$

$$= \int p_\theta(x) \sup_{t \in \text{dom } f^*} \{ tu - f^*(t) \} dx$$

$$= \int p_\theta(x) \sup_{t \in \text{dom } f^*} \left\{ t \left[\frac{p_x(x)}{p_\theta(x)} \right] - f^*(t) \right\} dx$$

Can't bring out sup directly as x is inside sup.

$$= \sup_{T(x) \in T^*} \int p_\theta(x) \left\{ T(x) \frac{p_x(x)}{p_\theta(x)} - f^*(T(x)) \right\} dx$$

The inner optimization problem involves x & the soln for it is dependent on it (a func of x)

$T : X \rightarrow \text{dom } f^*$ Space of func containing solutions for the inner opt. problem.

* we are search from a bag of func

*

$$D_f \geq \sup_{T(x) \in \mathcal{T}} \int p_0(x) \left\{ T(x) \frac{p_x(x)}{p_0(x)} - f^*(T(x)) \right\} dx$$

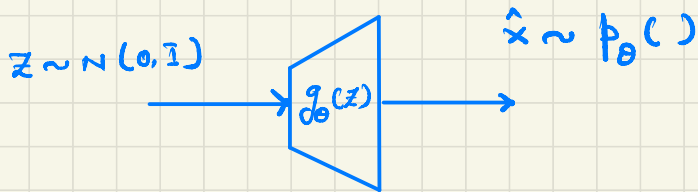
The space of func $\mathcal{T}$, that we are optimizing over may not contain the optimal $T^*(x)$ that is the solun for inner optimization.

$$\begin{aligned} D_f &\geq \sup_{T(x) \in \mathcal{T}} \int p_0(x) \left\{ T(x) \frac{p_x(x)}{p_0(x)} - f^*(T(x)) \right\} dx \\ &\geq \sup_{T(x)} \left[\int p_x(x) T(x) dx - \int p_0(x) f^*(T(x)) dx \right] \end{aligned}$$

$$D_f \geq \sup_{T(x)} \left[\mathbb{E}_{p_x} T(x) - \mathbb{E}_{p_0} f^*(T(x)) \right]$$

Realization of VDM:

Given Data $D = \{x_1, x_2, \dots, x_n\} \sim \text{iid } p_x$



$$\Theta^* = \underset{\Theta}{\operatorname{argmin}} D_b(p_x \parallel p_\theta)$$

$$D_b(p_x \parallel p_\theta) \geq \max_{T(x) \in \mathcal{T}} \left[\mathbb{E}_{p_x} T(x) - \mathbb{E}_{p_\theta} b^*(T(x)) \right]$$

$$\Theta^* = \underset{\Theta}{\operatorname{argmin}} D_b(p_x \parallel p_\theta)$$

$$\approx \underset{\Theta}{\operatorname{argmin}} \left[\text{Lower bound on } D_b \right]$$

We are optimizing the Lower bound.

$$= \underset{\Theta}{\operatorname{argmin}} \left[\max_{T(x)} \left(\mathbb{E}_{p_x} T(x) - \mathbb{E}_{p_\theta} b^*(T(x)) \right) \right]$$

Lower bound itself is an optimization problem.

Outer optimization: parameters of $g_\theta(z)$

Inner optimization

wrt a class of func

$$T(x) \in \mathcal{T}$$

Can't optimize over class of func

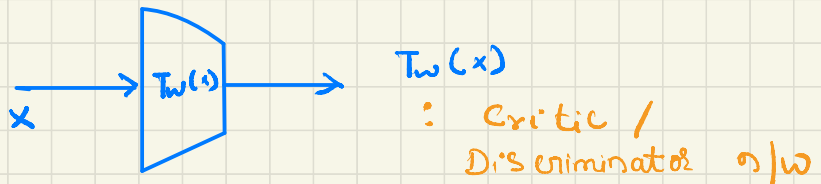
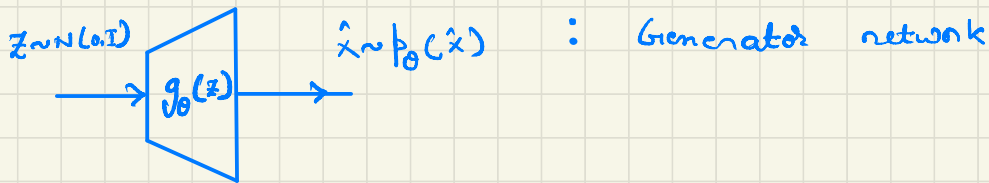
$\therefore$ Represent $\mathcal{T}$ via neural n/w $T_w(x)$

w : parameters of new n/w.

$\therefore$ now the objective will be

$$\Theta^*, w^* = \underset{\Theta}{\operatorname{argmin}} \max_w \left[\mathbb{E}_{p_x} T_w(x) - \mathbb{E}_{p_\theta} f^*(T_w(x)) \right]$$

Implementing VDM for generative Modeling



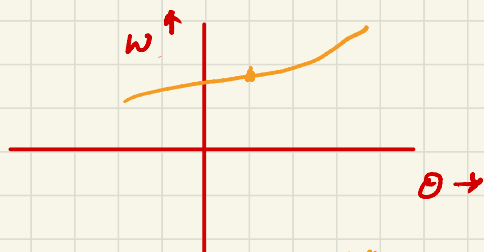
$$J(\theta, w) = \mathbb{E}_{p_x} T_w(x) - \mathbb{E}_{p_\theta} f^*(T_w(x))$$

$$\theta^*, w^* = \arg \min_{\theta} \max_w J(\theta, w)$$

alternating maximization & minimization problem.

its a saddle point problem.

⑤ Adversarial problem.



* at θ^*, w^* The func $J(\theta, w)$

increases around θ^* & decreases around w^*

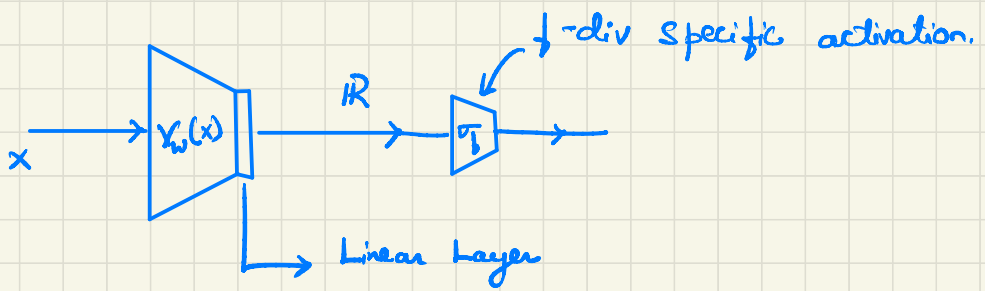
* difficult to optimize these kinds of problem.

$$T(\cdot) : X \rightarrow \text{dom } f^*$$

$$T_w(x) = \sigma_f(V_w(x)) \quad \text{where}$$

σ_f : f -divergence specific activation

$$V_w(x) : X \rightarrow \mathbb{R}$$



$$T_w(x) : X \rightarrow \text{dom } f^*$$

in practice

$$\underbrace{X \rightarrow \mathbb{R}}_{v_w(\cdot)} \rightarrow \underbrace{\text{dom } f^*}_{\sigma_f(\cdot)}$$

activation depends on the f -divergence under consideration.

$$J(\theta, w) = \mathbb{E}_{p_x} \left[\sigma_f(v_w(x)) \right] - \mathbb{E}_{p_\theta} \left[f^*(\sigma_f(v_w(x))) \right]$$

$$\sigma_f(w) : \mathbb{R} \rightarrow \text{dom } f^*$$